



REGULATIONS 2024
FLEXIBLE CHOICE BASED CREDIT SYSTEM
DEPARTMENT OF COMPUTER SCIENCE ENGINEERING
B.E. Computer Science and Engineering (IoT)
CURRICULUM - For Learners admitted from Academic Year 2024 - 2025

S.No	Remarks	Category	Course Code R2024	Course Code R2019	Course Title	Prerequisite	Theory Credits	Practical Credits	Total Credits	CGPA / Non - CGPA	Course Type
1	Any 1	HS	SH1104	19EN104	Technical Writing	-	3	0	3	CGPA	Theory
2	Any 1	HS	SH1105	19EN105	Public Speaking	-	0	3	3	CGPA	Practical (10 Expts)
3	Any 1	HS	SH1101	19EN101	Communicative English	-	3	1	4	CGPA	TIP (6 Expts)
4	Exempted for LE, Any 2 for Regular	HS	SH9154	19MS154	Basic Financial Accounting	-	3	0	3	CGPA	Theory
5	Exempted for LE, Any 2 for Regular	HS	SH9156	19MS156	Human Resource Management and Team Building	-	3	0	3	CGPA	Theory
6	Exempted for LE, Any 2 for Regular	HS	SH9155	19MS155	Stock Market and Company Operations	-	3	0	3	CGPA	Theory
7	Exempted for LE	HS	SH7101	22HS101	தமிழர் மரபு/ Heritage of Tamils	-	1	0	1	CGPA	Theory (Int Only)
8	Exempted for LE	HS	SH7102	22HS102	தமிழரும் தொழில்நுட்பமும்/ Tamils and Technology	-	1	0	1	CGPA	Theory (Int Only)
9	Exempted for LE	BS	SH7801	NA	Human Values and Professional Ethics	-	2	0	2	CGPA	Theory
10		BS	SH2222	19MA222	Probability and Queueing Models	-	3	1	4	CGPA	TIP (6 Expts)
11		BS	SH2212	19MA212	Algebra and Number Theory	19MA201	4	0	4	CGPA	Theory
12		BS	SH2220	19MA220	Mathematics for Artificial Intelligence	-	4	0	4	CGPA	Theory
13		BS	SH2221	19MA221	Linear Algebra Laboratory	-	0	2	2	CGPA	Practical (10 Expts)
14		BS	SH2211	19MA211	Statistics and Numerical Methods	-	3	1	4	CGPA	TIP (6 Expts)
15	Exempted for LE	BS	SH4205	19CY205	Principles of Chemistry in Engineering	-	3	1	4	CGPA	TIP (6 Expts)
16	Exempted for LE	ES	SH4801	NA	Environmental Sciences and Sustainability	-	2	0	2	CGPA	Theory
17	Exempted for LE	ES	SH3214	NA	Physics for Quantum Computing	-	2	1	3	CGPA	TIP (6 Expts)
18		ES	CS3301	19AI301	Python Programming	-	2	1	3	CGPA	TIP (6 Expts)
19	Exempted for LE	ES	CS3302	19AI302	Engineering Design and Modelling	-	0	3	3	CGPA	Practical (10 Expts)
20	Exempted for LE	ES	CS3303	19AI303	Engineering Mechanics and Product Development	-	2	1	3	CGPA	TIP (6 Expts)
21		ES	CS3306	19AI306	Object Oriented Programming using C++	-	2	1	3	CGPA	TIP (6 Expts)
22		ES	CS3307	19AI307	Object Oriented Programming using Java	19AI304	2	1	3	CGPA	TIP (6 Expts)
23		ES	CS3308	19AI308	Object Oriented Programming using C#	-	2	1	3	CGPA	TIP (6 Expts)
24	Exempted for LE	ES	EC2404	19EE404	Digital Electronics	-	3	1	4	CGPA	TIP (6 Expts)

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25	Exempted for LE	ES	EC2305	19EE305	Basic Electrical, Electronics and Measurement Engineering	-	2	1	3	CGPA	TIP (6 Expts)
26	Exempted for LE	ES	EC2309	19EE309	Programming Microcontrollers	-	2	1	3	CGPA	TIP (6 Expts)
27		PC	CS3304	19AI304	Fundamentals of C Programming	-	2	1	3	CGPA	TIP (6 Expts)
28		PC	CS3305	19AI305	Advanced C Programming	19AI304 / CS3304	2	1	3	CGPA	TIP (6 Expts)
29		PC	CS3414	19AI414	Fundamentals of Web Application Development	-	3	2	5	CGPA	TIP (10 Expts)
30	Exempted for LE	PC	CS1405	19CS405	Operating System	-	3	1	4	CGPA	TIP (6 Expts)
31	Exempted for LE	PC	CS1406	19CS406	Computer Networks	-	3	1	4	CGPA	TIP (6 Expts)
32		PC	CS1404	19CS404	Database Management System and its Applications	19AI301/19AI304/19CS301/19CS302/19CS303	3	1	4	CGPA	TIP (6 Expts)
33		PC	CS3408	19AI408	Data Structures	19AI304 / CS3304	2	1	3	CGPA	TIP (6 Expts)
34		PC	CS3404	19AI404	Analysis of Algorithms	19AI301	2	1	3	CGPA	TIP (6 Expts)
35		PC	CS1407	19CS407	Theory of Computation	19MA206	3	0	3	CGPA	Theory
36		PC	CS1408	19CS408	Software Engineering	-	3	1	4	CGPA	TIP (6 Expts)
37		PC	CS3405	19AI405	Fundamentals of Artificial Intelligence	19CS301	2	1	3	CGPA	TIP (6 Expts)
38		PC	CS3541	19AI541	Cloud Computing	19CS406	3	1	4	CGPA	TIP (6 Expts)
39		PC	CS1415	19CS415	Cryptography	-	2	1	3	CGPA	TIP (6 Expts)
40		PC	CS3547	19AI547	Blockchain for Business	19CS301	2	1	3	CGPA	TIP (6 Expts)
41		PC	CS1416	19CS416	Cloud Security	-	2	1	3	CGPA	TIP (6 Expts)
42		PC	CS1417	19CS417	Ethical Hacking Techniques	-	2	1	3	CGPA	TIP (6 Expts)
43		PC	CS1418	19CS418	Cyber Law and Compliance	-	3	0	3	CGPA	Theory
44		PC	CS1419	19CS419	Digital Forensics and Digital Investigations	-	2	1	3	CGPA	TIP (6 Expts)
45		PE	CS1565	19CS565	Robotic Process Automation	-	2	1	3	CGPA	TIP (6 Expts)
46		PE	CS1566	19CS566	Software Project Management and Version Control	-	3	0	3	CGPA	Theory
47		PE	CS1579	19CS579	Generative AI Application Development	-	2	1	3	CGPA	TIP (6 Expts)
48		PE	CS1568	19CS568	Fine Tuning Large Language Models	-	3	0	3	CGPA	Theory
49		PE	CS1569	19CS569	Applied Data Science	-	2	1	3	CGPA	TIP (6 Expts)
50		PE	CS1570	19CS570	Entrepreneurship and Small Business Development	-	2	1	3	CGPA	TIP (6 Expts)

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51		PE	CS1571	19CS571	Software Development and Operations	-	3	0	3	CGPA	Theory
52		PE	CS1572	19CS572	Serverless App Development	-	3	0	3	CGPA	Theory
53		PE	CS1573	19CS573	Enterprise Cloud Architecture	-	3	0	3	CGPA	Theory
54		PE	CS1574	19CS574	Advanced Class Design and Object-Oriented Techniques	-	3	0	3	CGPA	Theory
55		PE	CS1575	19CS575	Low-Code App Development	-	3	0	3	CGPA	Theory
56		PE	CS1577	19CS577	Cloud System Administration	-	3	0	3	CGPA	Theory
57		PE	CS1578	19CS578	Machine Learning Operations	-	3	0	3	CGPA	Theory
58		PE	CS1547	19CS547	Fundamentals of Cryptocurrency	-	3	0	3	CGPA	Theory
59		PE	CS1305	19CS305	Computer Architecture	-	3	0	3	CGPA	Theory
60		PE	CS1533	19CS533	Information Security Intelligence and Analytics using Big Data	-	2	1	3	CGPA	TIP (6 Expts)
61		PE	CS1534	19CS534	Prevention Methodology for Cybersecurity	-	2	1	3	CGPA	TIP (6 Expts)
62		PE	CS1535	19CS535	Trusted Network Systems	-	3	0	3	CGPA	Theory
63		PE	CS3526	19AI526	Information Security and Access Control	19AI527	2	1	3	CGPA	TIP (6 Expts)
64		PE	CS1536	19CS536	Firewalls and Intrusion Detection System	-	2	1	3	CGPA	TIP (6 Expts)
65		PE	CS3527	19AI527	Security Assessment and Risk Analysis	-	3	1	4	CGPA	TIP (6 Expts)
66		PE	CS3536	19AI536	Steganography and Digital Watermarking	-	2	1	3	CGPA	TIP (6 Expts)
67		PE	CS1537	19CS537	Secure Software Engineering	-	2	1	3	CGPA	TIP (6 Expts)
68		PE	CS1538	19CS538	System Security	-	3	0	3	CGPA	Theory
69		PE	CS1539	19CS539	Hardware Security	-	3	0	3	CGPA	Theory
70		PE	CS3409	19AI409	Applied Artificial Intelligence	19AI405/ 19CS301	2	1	3	CGPA	TIP (6 Expts)
71		PE	CS3501	19AI501	Applications of Artificial Intelligence	-	2	1	3	CGPA	TIP (6 Expts)
72		PE	CS3507	19AI507	Special Topics in Artificial Intelligence	-	2	1	3	CGPA	TIP (6 Expts)
73		PE	CS3503	19AI503	Computer Vision	19CS301	2	1	3	CGPA	TIP (6 Expts)
74		PE	CS3514	19AI514	Self Driving Car	19CS301/ 19AI410	2	2	4	CGPA	TIP (10 Expts)
75		PE	CS3522	19AI522	Knowledge Engineering	19CS410	3	0	3	CGPA	Theory
76		PE	CS3502	19AI502	Applied Natural Language Processing	-	2	1	3	CGPA	TIP (6 Expts)
77		PE	CS3410	19AI410	Introduction to Machine Learning	-	2	1	3	CGPA	TIP (6 Expts)
78		PE	CS3411	19AI411	Neural Networks	-	2	1	3	CGPA	TIP (6 Expts)
79		PE	CS3506	19AI506	Speech Analysis and Processing	-	2	1	3	CGPA	TIP (6 Expts)

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80		PE	CS3413	19AI413	Deep Learning and its Applications	-	2	1	3	CGPA	TIP (6 Expts)
81		PE	CS3521	19AI521	Expert Systems	-	3	0	3	CGPA	Theory
82		PE	CS3505	19AI505	Reinforcement Learning	19CS301	2	1	3	CGPA	TIP (6 Expts)
83		PE	CS4501	19AM501	Machine Learning for Bioinformatics	-	3	0	3	CGPA	Theory
84		PE	CS4502	19AM502	Genetic Algorithms	-	2	1	3	CGPA	TIP (6 Expts)
85		PE	CS4505	19AM505	Statistical Learning Theory	-	2	1	3	CGPA	TIP (6 Expts)
86		PE	CS3533	19AI533	Introduction to Robotics	-	2	1	3	CGPA	TIP (6 Expts)
87		PE	CS3534	19AI534	Kinematics and Dynamics of Robots	-	2	1	3	CGPA	TIP (6 Expts)
88		PE	CS3535	19AI535	Robotic Sensors	-	2	1	3	CGPA	TIP (6 Expts)
89		PE	CS3530	19AI530	Control of Robotic Systems	-	2	1	3	CGPA	TIP (6 Expts)
90		PE	CS3509	19AI509	Concepts of Virtual And Augmented Reality	-	2	1	3	CGPA	TIP (6 Expts)
91		PE	CS3510	19AI510	Mobile VR And AI In Unity	19AI410/ 19CS302	2	1	3	CGPA	TIP (6 Expts)
92		PE	CS3513	19AI513	Game Programming	19CS401	2	2	4	CGPA	TIP (10 Expts)
93		PE	CS3537	19AI537	Computer Graphics for Virtual Reality	-	2	1	3	CGPA	TIP (6 Expts)
94		PE	CS4514	19AM514	Motion Planning Techniques	-	2	1	3	CGPA	TIP (6 Expts)
95		PE	CS3548	19AI548	Robot Programming	-	2	1	3	CGPA	TIP (6 Expts)
96		PE	CS3549	19AI549	Drone Assembly and Applications	-	2	1	3	CGPA	TIP (6 Expts)
97		PE	CS3550	19AI550	Artificial Intelligence for Games	-	2	1	3	CGPA	TIP (6 Expts)
98		PE	CS3551	19AI551	Drone Design and Control	-	2	1	3	CGPA	TIP (6 Expts)
99		PE	CS3539	19AI539	Mobile User Interface Development	19CS401	2	1	3	CGPA	TIP (6 Expts)
100		PE	CS3540	19AI540	Programming Mobile Devices	19CS401	2	1	3	CGPA	TIP (6 Expts)
101		PE	CS3545	19AI545	Modern Web Application Development	19CS410	2	1	3	CGPA	TIP (6 Expts)
102		PE	CS3546	19AI546	Web Server Programming	19CS401	2	1	3	CGPA	TIP (6 Expts)
103		PE	CS3552	19AI552	Web Application User Interface Development	-	2	1	3	CGPA	TIP (6 Expts)
104		PE	CS3553	19AI553	Advanced Java Web Applications	19AI307	2	1	3	CGPA	TIP (6 Expts)
105		PE	CS3403	19AI403	Introduction to Data Science	19CS301	2	1	3	CGPA	TIP (6 Expts)
106		PE	CS3511	19AI511	Scientific and Engineering Data Visualization	-	2	1	3	CGPA	TIP (6 Expts)
107		PE	CS3516	19AI516	Big Data Mining and Analysis	19AI408/ 19CS410/ 19IT405	2	1	3	CGPA	TIP (6 Expts)
108		PE	CS3517	19AI517	Business Analytics	-	2	1	3	CGPA	TIP (6 Expts)

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109		PE	CS4503	19AM503	Data Modeling	-	3	0	3	CGPA	Theory
110		PE	CS4508	19AM508	Introduction to IoT	-	2	1	3	CGPA	TIP (6 Expts)
111		PE	CS1420	19CS420	Prototyping of IoT Systems	-	2	1	3	CGPA	TIP (6 Expts)
112		PE	CS1421	19CS421	Edge Computing	-	2	1	3	CGPA	TIP (6 Expts)
113		PE	CS4509	19AM509	Industrial Internet of Things	-	2	1	3	CGPA	TIP (6 Expts)
114		PE	CS1422	19CS422	Security for IoT Systems	-	3	0	3	CGPA	Theory
115		PE	NA	19EC508	Ad Hoc and Wireless Sensor Networks	19EC412	3	0	3	CGPA	Theory
116		PE	CS1540	19CS540	Fog Computing	-	3	0	3	CGPA	Theory
117		PE	CS4506	19AM506	Sensors and Actuators for IoT	-	2	1	3	CGPA	TIP (6 Expts)
118		PE	CS1542	19CS542	Embedded Board Design	-	2	1	3	CGPA	TIP (6 Expts)
119		PE	CS1543	19CS543	Embedded Board Manufacturing	-	2	1	3	CGPA	TIP (6 Expts)
120		PE	CS1544	19CS544	Realtime Operating System	-	2	1	3	CGPA	TIP (6 Expts)
121		PE	CS1545	19CS545	Open Source Operating System	-	2	1	3	CGPA	TIP (6 Expts)
122		PE	CS1546	19CS546	Kernel Programming	-	2	1	3	CGPA	TIP (6 Expts)
123		PE	CS3538	19AI538	Mathematical Modelling and Computer Aided Engineering	-	2	1	3	CGPA	TIP (6 Expts)
124		PE	CS3542	19AI542	Agile Software Development	19CS301	3	1	4	CGPA	TIP (6 Expts)
125		PE	CS3543	19AI543	Software Testing	19CS301	2	1	3	CGPA	TIP (6 Expts)
126		PE	CS3544	19AI544	Virtualization and Containerization	19CS405/ 19CS406	2	1	3	CGPA	TIP (6 Expts)
127		PE	CS3528	19AI528	Advanced Graph Theory and Applications	19CS402	2	1	3	CGPA	TIP (6 Expts)
128		PE	CS3529	19AI529	Advanced Data Structures	19CS401	2	1	3	CGPA	TIP (6 Expts)
129		PE	CS3508	19AI508	Soft Computing Techniques	-	2	1	3	CGPA	TIP (6 Expts)
130		PE	CS3515	19AI515	Smart Manufacturing Technology	-	2	1	3	CGPA	TIP (6 Expts)
131		PE	CS1523	19CS523	Information Retrieval	19AI410/ 19CS501/19 CS413	3	0	3	CGPA	Theory
132		PE	CS3512	19AI512	NoSQL Database Design	19CS301	2	1	3	CGPA	TIP (6 Expts)
133		PE	CS3518	19AI518	Cognitive Systems	-	3	0	3	CGPA	Theory
134		PE	CS3519	19AI519	Distributed Database	19CS404	3	0	3	CGPA	Theory
135		PE	CS3520	19AI520	Data Mining and Warehousing	-	2	1	3	CGPA	TIP (6 Expts)
136		PE	CS3523	19AI523	Mobile Database	-	3	0	3	CGPA	Theory
137		PE	CS3524	19AI524	Multimedia Database	19CS410	3	0	3	CGPA	Theory
138		PE	CS3525	19AI525	Video Processing	19CS301	2	1	3	CGPA	TIP (6 Expts)
139		PE	CS3406	19AI406	Digital Image Processing Techniques	-	2	1	3	CGPA	TIP (6 Expts)

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140		PE	CS3407	19AI407	Parallel Computing Architecture	-	3	1	4	CGPA	TIP (6 Expts)
141		PE	CS3412	19AI412	Web Data Mining	-	3	1	4	CGPA	TIP (6 Expts)
142		OE	XXXXXX	19XXXXX	Open Electives	-	0	0	0	CGPA	Theory
143		OE	XXXXXX	19XXXXX	Online Courses	-	0	0	0	CGPA	Experiential Learning
144		EEC	CS3701	19AI701	Mini Project	-	0	2	2	CGPA	Project (Int Only)
145	Option 1	EEC	CS3702	19AI702	Project Work I	-	0	2	2	CGPA	Practical (10 Expts)
146	Option 1	EEC	CS3703	19AI703	Project Work II	19AI702	0	6	6	CGPA	Project
147	Option 2	EEC	SH6802	NA	Start Up	-	0	8	8	CGPA	Project
148		EEC	SH6708	19EY708	Career Development Skills	NIL	1	0	1	CGPA	Theory (Int Only)
149		EEC	SH6709	19EY709	Reasoning Ability	NIL	1	0	1	CGPA	Theory (Int Only)
150		EEC	SH6710	19EY710	Quantitative Ability I	NIL	1	0	1	CGPA	Theory (Int Only)
151		EEC	SH6711	19EY711	Quantitative Ability II	SH6710 / 19EY710	1	0	1	CGPA	Theory (Int Only)
152		EEC	SH6712	19EY712	Employment Enrichment Skills	19EY702 / 19EY709 / SH6709	1	0	1	CGPA	Theory (Int Only)
153	Any 1	EEC	NA	19EY706	Company Specific Assessments For Employability	-	0	1	1	CGPA	Practical (6 Expts)
154	Any 1	EEC	SH6713	19EY713	Placement Readiness Assessment	19EY704 / 19EY711 / SH6711	1	0	1	CGPA	Theory (Int Only)
155		EEC	SH6707	19EY707	Paper Publication	-	0	0	1	NON-CGPA	Experiential Learning (Int Only)
156		MC	SH8801	NA	Constitutionalism and Artificial Intelligence Policies in India	-	1	0	1	NON-CGPA	Theory (Int Only)
157	Any 1	MC	SH8802	NA	Internship	-	0	0	2	NON-CGPA	Experiential Learning (Int Only)
158	Any 1	MC	SH8803	NA	Entrepreneurship	-	0	0	2	NON-CGPA	Experiential Learning (Int Only)
159	Any 1	MC	SH6803	19MC807	National Service Scheme	-	0	0	0	NON-CGPA	Experiential Learning (Int Only)
160	Any 1	MC	SH8807	19MC808	National Sports Organization	-	0	0	0	NON-CGPA	Experiential Learning (Int Only)
161	Any 1	MC	SH8808	19MC809	Youth Red Cross	-	0	0	0	NON-CGPA	Experiential Learning (Int Only)

BATCH: 2024 - 2028 Onwards		
Category	Credits to be Earned (Regular)	Credits to be Earned (Lateral Entry)
Humanities and Science	14	3
Basic Science	25	18
Engineering Science	28	12
Professional Core	56	47
Professional Electives	16	16
Open Electives	12	12
Employability Enhancement Courses	16	16
Mandatory Courses	4	4
TOTAL	171	128